

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Ethiopian Intercity/Ride-Hail Commuter Cohort (Amharic-Primary)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotation methodology is sound on its face: ≥ 3 native annotators per utterance with majority voting [\[Q42, Q45\]](#), Fleiss' Kappa improving to 0.912–1.00 after a two-round review [\[Q47, Q48\]](#), with Amharic specifically reaching $\kappa=0.933$ post-review (initial 0.836). However, several documented and empirical gaps reduce convergent validity for this deployment: (a) annotator regional/L2 demographics within Ethiopia are undisclosed [\[Q44, gap-4 NOT FOUND\]](#); (b) the actual user population includes L2 Amharic speakers from Oromia, Tigray, and SNNP regions who code-switch — a population mismatch flagged HIGH in elicitation; (c) dataset inspection reveals concrete annotation gaps (empty spans on utterances containing clear DATE/TIME content [\[DATASET-D11, D33\]](#), inconsistent treatment of '12 ሰዓት' across alarm utterances [\[DATASET-D11 vs D23\]](#), misannotated PERSONAL_NAME in Oromo [\[DATASET-D29\]](#)); (d) no code-switched utterances at all in 19 Amharic samples [\[DATASET-D1–D18\]](#). OC was marked HIGH priority in elicitation, so these residuals weigh heavily.

ID	Evidence
Q42	"Each utterance was annotated by three annotators so that we could check for agreement in the slot annotations." (p.4)
Q44	"For both utterance elicitation and slot-filling annotation, all recruited participants received an appropriate remuneration based on the per-country rate decided by our logistics company in Kenya." (p.4)
Q45	"To ensure annotation quality and consistency, we follow a rigorous quality control process using a majority voting system with a minimum of three annotators per sentence to resolve disagreements." (p.4)
Q47	"Initial Fleiss' Kappa scores revealed substantial variation across languages, ranging from 0.618 (Zulu) to 0.934 (Shona), indicating significant inter-annotator disagreement." (p.4)
Q48	"Following the review process, agreement scores improved markedly across all languages, reaching 0.912-1.00." (p.4)
D1	ከድሬዳዋ ወደ ባህርዳር ለመስከረም አምስት በረራ ያዝልኝ። — "Book me a flight from Dire Dawa to Bahir Dar for Meskerem fifth."
D2	target: "CITY_OR_PROVINCE: ድሬዳዋ \$\$\$ CITY_OR_PROVINCE: ባህርዳር \$\$\$ DATE: መስከረም አምስት" — Slot annotations for Dire Dawa, Bahir Dar (Ethiopian cities), and Meskerem (Ethiopian calendar month)
D3	የአማራ ባንክ አካውንቱ ስንት አለው? / target: "BANK_NAME: አማራ ባንክ — "How much is in my Amhara Bank account?" — Ethiopian bank name annotated
D4	ንግድ ባንክ ላይ የቀረኝ ገንዘብ አለ? / target: "BANK_NAME: ንግድ ባንክ — "Is there money left in Commercial Bank [of Ethiopia]?" — major Ethiopian bank
D5	በቴሌብር የብድር አካውንቱ ላይ ... / target: "PAYMENT_COMPANY: ቴሌብር — "In my TeleBirr credit account..." — TeleBirr is Ethiopia's dominant mobile payment platform
D6	ከሃምሌ 4 እስከ 6 የቤት መኪና ኪራይ ማግኘት እችላለሁ? / target: "DATE: ከሃምሌ 4 እስከ 6 — "Can I find a car rental from Hamle 4 to 6?" — Hamle is the 11th month of Ethiopian calendar
D7	ወደ ቱርክ በፍጥነት የምድርስበት የበረራ አማራጭ ንግረኝ? / target: "COUNTRY: ቱርክ — "Tell me a flight option to get to Turkey quickly." — international destination, no Ethiopian intercity booking
D8	ወደ ሱዳን የበረራ ትኬት ያስፈልገኛል። / target: "COUNTRY: ሱዳን — "I need a flight ticket to Sudan." — international destination, not intercity bus
D9	raw: "ለቀጣይ የጉብኝት ጉዞዬ በሃዋሳ እና በአርባ ምንጭ መካከል..." / text: "help me decide between Hawassa and Arbaminch..." — Ethiopian cities Hawassa and Arba Minch appear as travel suggestions — confirms cultural grounding
D10	raw: "ከቴሌብር አካውንቱ ገንዘብ ለማን እንደላኩ ዝርዝሩን አሳየኝ?" / text: "Show me history of my telebirr transfers?" — TeleBirr payment company referenced in Amharic utterance

D11	ነገ ጠዋት 12 ሰአት የቀጠርኩትን ማንቂያ በደንብ እንዲሰማኝ ድምጽ ጨምርበት — "Make the alarm I set for tomorrow morning at 12 o'clock ring loudly." — TIME slot unannotated despite "12 ሰአት"
D12	Balallii Dirree dawaa irraa gara Boolee achiin gara jimmaa Xiyyaara Itoopiyya irraa naaf kuti — "Book me an Ethiopian Airlines flight from Dire Dawa to Bole then to Jimma." — spans empty despite city names and airline
D13	Ebila 12 hanga Ebila 14tti Mattuu keessatti kireeffannaa konkolaataa argachuun ni danda'amaa" / target: "DATE: Ebila 12 hanga Ebila 14 \$\$\$ CITY_OR_PROVINCE: Mattuu — Oromo month name "Ebila" (April) used as date slot — culturally grounded Ethiopian Oromo calendar
D14	Sagantaa sirna cidhaaf sadaasa 21tti qabame kalaandarii koo irraa haqi maloo" / target: "CALENDAR_EVENT: sirna cidhaa \$\$\$ DATE: sadaasa 21 — Oromo month "Sadaasa" (November in Oromo calendar) used as date slot
D15	የዚህ ወር መከፈል ያለባቸውን የፍጆታ ዝርዝር አሳየኝ::" / target: "DATE: የዚህ ወር — "Show me the utility list to be paid this month." — relative date expression, not calendar month name
D16	ከሚመጣው ወር ጀምሮ ለ3 ወር ቶዮታ መኪና መከራየት እፈልጋለሁ::" / target: "DATE: ከሚመጣው ወር ጀምሮ ለ3 ወር — "I want to rent a Toyota car starting next month for 3 months." — relative date, no named calendar month
D17	በቶቶ ለቃለአብ ቦታ ማስያዝ እንዳለብኝ..." / target: "RESTAURANT_NAME: ቶቶ \$\$\$ PERSONAL_NAME: ቃለአብ — Restaurant "ቶቶ" (Totot) — Ethiopian cultural reference; personal name "ቃለአብ" is distinctly Ethiopian
D18	የቦሎ ማሳደሻ ስንት ነው" / target: "BILL_TYPE: የቦሎ ማሳደሻ — "What is the Bole maintenance/upgrade fee?" — "ቦሎ" likely refers to Bole (Addis Ababa district)
D23	"ነገ ጠዋት 12 ሰአት..." (no span annotation) — "12 ሰአት" (12 o'clock) unannotated — "ጠዋት" (morning) absent from target; Ethiopian time convention unclear
D29	Galgala kana caalaan akka dhufuuf dammaksituu guutiif saatii 7tti" / target includes misannotated "PERSONAL_NAME: caalaa — "caalaa" annotated as PERSONAL_NAME but may be part of "Galgala caalaan" (this evening) phrase
D33	የእናቴን እና የአባቴን የልደት ቀን የቀን መቁጠሪያዬ ላይ ስላቀያየርኩት አስተካክለው — "I changed my parents' birthdays in my calendar, please fix it." — no slots annotated despite calendar events

Strengths

- Native-speaker annotation with three annotators and majority voting [Q42, Q45]
- Two-round review process raised inter-annotator agreement substantially [Q47, Q48, Q109]
- Amharic post-review $\kappa=0.933$, second-highest among all initial scores [WEB-17]
- Detailed annotation guide with edge-case rules [Q149, Q156, Q157]

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Q42	"Each utterance was annotated by three annotators so that we could check for agreement in the slot annotations." (p.4)
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Q47	"Initial Fleiss' Kappa scores revealed substantial variation across languages, ranging from 0.618 (Zulu) to 0.934 (Shona), indicating significant inter-annotator disagreement." (p.4)
Q48	"Following the review process, agreement scores improved markedly across all languages, reaching 0.912-1.00." (p.4)
Q109	"The annotation process revealed inherent challenges in entity classification across languages, requiring two rounds of review to achieve consistent quality." (p.9)
Q149	"This section provides the complete annotation guide and instruction for annotators working for labeling all slots types." (p.20)
Q156	"With this entity, only annotate if entity is named explicitly, e.g Name of airport, museum or mall is not and nt just "mall", "airport" etc" (p.24)
Q157	"PS: Do not skip any annotations, if there is nothing to annotate, submit and go to the next one." (p.24)
WEB-17	Amharic post-review $\kappa=0.933$ supports output reliability (https://www.researchgate.net/publication/389056008_INJONGO_A_Multicultural_Intent_Detection_and_Slot-filling_Dataset_for_16_African_Languages)

Checklist

Checklist item definitions:

ID	Assessment Question
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OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Partially. Native Amharic annotators were used [Q7], but their regional/L2 representativeness for the actual deployment user base (multi-regional, L2-heavy) is undisclosed [Q44].

OC-2: Likely disagreement on (a) code-switched utterances (absent from training entirely), (b) Ethiopian time convention, (c) regional toponyms beyond major cities. Demographic profile of annotators is not published.

OC-3: Documentation specifies 'native speakers' [Q7] and recruitment via a Kenyan logistics company [Q44]. No demographic breakdown by region, urban/rural, age, or L1/L2 status is provided.

OC-4: Re-annotation by a representative pool (including non-Addis L2 speakers and code-switching annotators) is recommended given the deployment's user heterogeneity.

OC-5: Majority voting [Q45] is used. With three annotators and absence of demographic diversity controls, minority regional perspectives may be erased; the review process explicitly resolved disagreement toward consistency [Q53, Q109].

OC-6: Documented issues: undisclosed demographics, observed annotation gaps in Amharic [DATASET-D11, D33], absence of code-switched data — all harming convergent and external validity.

ID	Evidence
Q7	"utterances generated by native speakers across diverse domains, including banking, travel, home, and dining." (p.1)
Q44	"For both utterance elicitation and slot-filling annotation, all recruited participants received an appropriate remuneration based on the per-country rate decided by our logistics company in Kenya." (p.4)
Q45	"To ensure annotation quality and consistency, we follow a rigorous quality control process using a majority voting system with a minimum of three annotators per sentence to resolve disagreements." (p.4)
Q53	"Examination of annotator feedback and comparative analysis between unreviewed and reviewed versions indicated that ambiguous slot types significantly impacted annotation quality and introduced unnecessary complexity." (p.4)
Q109	"The annotation process revealed inherent challenges in entity classification across languages, requiring two rounds of review to achieve consistent quality." (p.9)
D11	ነገ ጠዋት 12 ሰዓት የቀጠርኩትን ማንቂያ በደንብ እንዲሰማኝ ድምጽ ጨምርበት — "Make the alarm I set for tomorrow morning at 12 o'clock ring loudly." — TIME slot unannotated despite "12 ሰዓት"
D33	የእናቴን እና የአባቴን የልደት ቀን የቀን መቁጠሪያዬ ላይ ስላቀያየርኩት አስተካክለው — "I changed my parents' birthdays in my calendar, please fix it." — no slots annotated despite calendar events
WEB-3	Amharic L2 lingua-franca status — most learned second language across Ethiopia (https://arxiv.org/html/2603.23654)

Information Gaps

- Annotator demographics within Ethiopia (region of origin, urban/rural, L1/L2 Amharic, age, education)
- Number of annotators recruited for the Amharic split specifically
- Whether any annotation review was performed by Ethiopian regional sub-population

Requires Expert Verification

- Native Amharic speaker review of suspected misannotations and empty-span examples to confirm rate of annotation gaps

Remediation

Gap 1: Annotator regional/L2 demographics within Ethiopia undisclosed; observed annotation gaps (empty spans on utterances with clear DATE/TIME content); no code-switched data ever annotated

Recommendation: Recruit a regionally diverse annotator pool — explicitly including non-Addis annotators from Amhara, Oromia, Tigray, SNNP, and Somali regions, with documented L1/L2 status — and re-annotate a stratified sample (~500 utterances) of the Amharic INJONGO test split plus a new code-switched mini-set; report κ stratified by annotator subgroup.